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Roll No.

**B.TECH.- THIRD SEMESTER
END-SEMESTER EXAMINATION, November-December
2025**

Course Code - EENTC04/ICNITC04
Course Title - Mathematics III

Time: 3:00 hours

Max Marks-50

Note: Attempt all questions in given order only. Missing data/information (if any), maybe suitably assumed by mentioned in the answer.

Q. No.	Question	Marks	CO
1(a)	State and prove Cauchy's integration formula.	5	CO1
1(b)	Find the analytic function whose real part is $u(x, y) = x^3 - 3xy^2 + 3x^2 - 3y^2$	5	CO1
1(c)	Determine the residue of $f(z) = \frac{z^4}{(z-1)^4(z-2)(z-3)}$ at its simple poles.	5	CO1
2(a)	State and prove Green's theorem.	5	CO2
2(b)	Verify divergence theorem for $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ taken over the cube bounded by the planes $x = 0$, $x = 1$, $y = 0$, $y = 1$, $z = 0$, $z = 1$.	5	CO2
2(c)	Use Stoke's theorem to evaluate $\int_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = x^2\hat{i} + y^2\hat{j} + x\hat{k}$ and C is the triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$ with counter-clockwise rotation.	5	CO2
3(a)	State convolution theorem and by using it find the inverse Laplace transform of $\frac{1}{(s+a)(s^2+1)}$	5	CO3

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3(b)	Solve the differential equation $y'' + 3y' + 2y = te^{-t}$ with $y(0) = 1$ and $y'(0) = 0$.	5	CO3														
3(c)	Find the Laplace transform of the following functions: (i) $f(t) = te^{-at} \sin 3t$, (ii) $f(t) = \frac{1 - \cos t}{t}$.	5	CO3														
4(a)	Find the Fourier series expansion of the function $f(x) = \frac{x-x^2}{2}$ in the interval $(0, 2\pi)$ and deduce that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$	5	CO4														
4(b)	Find the half range cosine series of the function $f(x) = \sin \frac{\pi x}{2}$ in the interval $0 < x < l$.	5	CO4														
4(c)	Compute the Fourier coefficients a_0, a_1, a_2, a_3 in the expansion of the function tabulated as follows: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>4</td><td>8</td><td>15</td><td>7</td><td>6</td><td>2</td></tr></table>	x	0	1	2	3	4	5	y	4	8	15	7	6	2	5	CO4
x	0	1	2	3	4	5											
y	4	8	15	7	6	2											
5(a)	Find the solution of the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ which satisfies the following boundary conditions: (i) $u \rightarrow 0$ as $y \rightarrow \infty$ for all x , (ii) $u = 0$ at $x = 0$ for all y , (iii) $u = 0$ at $x = 1$ for all y , (iv) $u = x(1 - x)$ if $y = 0$ for all $x \in (0, 1)$.	5	CO5														
5(b)	Solve the heat equation $\frac{\partial u}{\partial t} = k^2 \frac{\partial^2 u}{\partial x^2}$, if $u(0, t) = u(l, t) = 0$ and $u(x, 0) = 50$.	5	CO5														
5(c)	A string is attached and fastened to two points distance l apart and displaced into the form $u(x, 0) = 3(lx - x^2)$ from which it is released at time $t = 0$. Find the displacement $u(x, t)$ of the string at a distance x from one end.	5	CO5														